



Calculus & Statistics

Tutorial-8

Triple integration & Volume by triple Integration

Q1. Evaluate the following triple integrals :

- 1) $\int_{-c}^c \int_{-b}^b \int_{-a}^a (x^2 + y^2 + z^2) dx dy dz$
- 2) $\int_0^{\pi/2} \int_0^{a \sin \theta} \int_0^{\frac{a^2 - r^2}{a}} r dz dr d\theta$
- 3) $\int_1^e \int_1^{\log y} \int_1^{e^x} \log z \, dx dy dz$
- 4) $\int_0^2 \int_1^z \int_0^{yz} xyz \, dx dy dz$
- 5) $\int_0^1 \int_{y^2}^1 \int_0^{1-x} x dz dx dy$
- 6) $\iiint z^2 \, dx dy dz$ over the region common to the sphere $x^2 + y^2 + z^2 = 4$ & the cylinder $x^2 + y^2 = 2x$.
- 7) $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{dx dy dz}{\sqrt{1-x^2-y^2-z^2}}$ using spherical polar coordinates
- 8) $\iiint z^2 \, dx dy dz$ over the volume bounded by the surfaces $x^2 + y^2 = a^2$, $x^2 + y^2 = z$ and $z = 0$
- 9) $\iiint \frac{dx dy dz}{(1+x+y+z)^3}$ over the volume of the tetrahedron $x = 0, y = 0, z = 0, x + y + z = 1$
- 10) $\iiint (x + y + z) dx dy dz$ over the tetrahedron bounded by the planes $x=0, y=0, z=0$ and $x + y + z = 1$.

Q2. Find the volume of the given solid in the following problems:

- 1) Find the volume of the tetrahedron bounded by the planes $x = 0, y = 0, z = 0$ & $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.
- 2) Find the volume of the solid bounded by the paraboloids $z = x^2 + y^2$ & $z = 4 - 3(x^2 + y^2)$.
- 3) Find the volume bounded by the cylinder $x^2 + y^2 = 4$ & the planes $y + z = 4$ and $z = 0$
- 4) Find the volume bounded by the sphere $x^2 + y^2 + z^2 = 9$
- 5) Find the portion of the sphere $x^2 + y^2 + z^2 = a^2$ lying inside the cylinder $x^2 + y^2 = ay$